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Rotary flow control valve that requires no linear motion

Abstract

A rotary flow control valve that requires no linear motion is disclosed. The rotary flow control valve comprises a drive mechanism and a rotary flow control valve coupled with the drive mechanism, wherein the rotary flow control valve is rotated by the drive mechanism without requiring conversion of a linear motion to a rotational motion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (PROVISIONAL) (1) This application claims priority to and benefit of U.S. Provisional Patent Application No. 63/146,399 filed on Feb. 5, 2021, entitled “Rotary Valve For Dropper Seatpost” by Joshua Coaplen et al., and assigned to the assignee of the present application, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

(1) Embodiments of the invention generally relate to systems and methods for actuating hydraulic flow states in a dropper seatpost.

BACKGROUND

(2) Prior rigid seatpost designs have begun to be replaced with dropper seatpost assemblies. In a dropper seatpost assembly, the seatpost can be lowered or raised by a rider via a lever. When the rider operates the lever, the dropper seatpost will move the saddle from the riders set saddle height to a lowered position, e.g., moving the saddle down and out of the rider's way. The rider can then operate the lever again and the dropper seatpost will return the saddle to the riders previously established saddle height. Often, however, dropper seatpost assemblies can add undesired amounts of weight, complexity, and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present invention are illustrated by way of example, and not by way of limitation, in the accompanying drawings, wherein:

(2) FIG. 1 is a perspective view of a bicycle, in accordance with an embodiment.

(3) FIG. 2 is a perspective view of a handlebar with a control lever coupled therewith, in accordance with an embodiment.

(4) FIG. 3A is a perspective view of a rotary flow control valve in a dropper seatpost assembly, in accordance with an embodiment.

(5) FIG. 3B is a cross-section view of the rotary flow control valve in a dropper seatpost assembly, in accordance with an embodiment.

(6) FIG. 4A is a cross-section view (identified in FIG. 3B) of a portion of the dropper seatpost assembly including a motor to electronically drive the rotary flow control valve assembly, in accordance with an embodiment.

(7) FIG. 4B is a cross-section view (identified in FIG. 3B) of a portion of the dropper seatpost assembly including a mechanical actuator for rotary flow control valve assembly, in accordance with an embodiment.

(8) FIG. 5A is a perspective view of a rotary flow control valve, in accordance with an embodiment.

(9) FIG. 5B is a partial cutaway view of the rotary flow control valve of FIG. 5A installed within the rotary flow control valve assembly in an open position, in accordance with an embodiment.

(10) FIG. 5C is a cross-section view of the rotary flow control valve of FIG. 5A installed within the rotary flow control valve assembly in an open position, in accordance with an embodiment.

(11) FIG. 5D is a partial cut-away view of the rotary flow control valve of FIG. 5A installed within the rotary flow control valve assembly in a closed position, in accordance with an embodiment.

(12) FIG. 5E is a cross-section view of the rotary flow control valve of FIG. 5A installed within the rotary flow control valve assembly in a closed position, in accordance with an embodiment.

(13) FIG. 6A is a perspective view of the head of rotary flow control valve with one or more O-ring type seals installed in a gland, in accordance with an embodiment.

(14) FIG. 6B is a top view of the head of rotary flow control valve with one or more O-ring type seals installed in glands thereon, in accordance with an embodiment.

(15) FIG. 7A is a perspective cross-section view of the rotary flow control valve with one or more preload pads installed thereon, in accordance with an embodiment.

(16) FIG. 7B is a perspective view of the rotary flow control valve of FIG. 7A with one or more preload pads installed thereon in accordance with an embodiment.

(17) FIG. 7C is a side view of the rotary flow control valve of FIG. 7A with one or more preload

pads installed thereon in accordance with an embodiment.

(18) FIG. 8A is a top perspective cross-section view of a rotary flow control valve with one or more preload pads installed thereon, in accordance with an embodiment.

(19) FIG. 8B is a perspective view of the rotary flow control valve of FIG. 8A with one or more preload pads installed thereon, in accordance with an embodiment.

(20) FIG. 9A is a perspective view of a rotary flow control valve with an optional flow hole therethrough, in accordance with an embodiment.

(21) FIG. 9B is a cross-section view of the rotary flow control valve of FIG. 9A with a flow hole therethrough installed within the rotary flow control valve assembly, in accordance with an embodiment.

(22) FIG. 9C is a top perspective cross-section view of a rotary flow control valve with a flow hole therethrough installed within the rotary flow control valve assembly, in accordance with an embodiment.

(23) FIG. 9D is a cross-section view of the rotary flow control valve with a flow hole therethrough installed within the rotary flow control valve assembly, in accordance with an embodiment.

(24) FIG. 10A is a cross-section view of the rotary flow control valve with a single O-ring configuration installed within the rotary flow control valve assembly, in accordance with an embodiment.

(25) FIG. 10B is a perspective view of the rotary flow control valve with a single O-ring configuration of FIG. 10A, in accordance with an embodiment.

(26) FIG. 10C is a top perspective cut away view of the rotary flow control valve with a single O-ring configuration installed within the rotary flow control valve assembly in a closed state, in accordance with an embodiment.

(27) FIG. 10D is a top perspective cut away view of the rotary flow control valve with a single O-ring configuration installed within the rotary flow control valve assembly in an open state, in accordance with an embodiment.

(28) FIG. 11A is a cross-section view of a portion of a rotary flow control valve assembly with a rotating shaft configuration, in accordance with an embodiment.

(29) FIG. 11B is a top cross-section view of the rotary flow control valve assembly with a rotating drive shaft in a closed state configuration, in accordance with an embodiment.

(30) FIG. 11C is a top cross-section view of the rotary flow control valve assembly with a rotating drive shaft in an open state configuration, in accordance with an embodiment.

(31) FIG. 11D is a cross-section view of a portion of a rotary flow control valve assembly with a rotating shaft configuration having a one piece ball valve seat, in accordance with an embodiment.

(32) FIG. 12A is a cross-section view of a portion of a rotary flow control valve assembly with a rotating drive shaft in the form of a cylindrical shaft configuration, in accordance with an embodiment.

(33) FIG. 12B is a top cross-section view of the rotary flow control valve assembly with a rotating drive shaft in the form of a cylindrical shaft configuration in a closed state, in accordance with an embodiment.

(34) FIG. 12C is a top cross-section view of the rotary flow control valve assembly with a rotating drive shaft in the form of a cylindrical shaft configuration in an open state, in accordance with an embodiment.

(35) FIG. 13A is a cross-section view of a portion of a dual O-ring rotary flow control valve assembly with a rotating drive shaft in the form of a cylindrical shaft configuration, in accordance with an embodiment.

(36) FIG. 13B is a top cross-section view of the dual O-ring rotary flow control valve assembly with a rotating drive shaft in the form of a cylindrical shaft configuration in a closed state, in accordance with an embodiment.

(37) FIG. 13C is a top cross-section view of the dual O-ring rotary flow control valve assembly

with a rotating drive shaft in the form of a cylindrical shaft configuration in an open state, in accordance with an embodiment.

(38) FIG. 14A is a perspective view of a rotating drive shaft in the form of a cylindrical shaft configuration with one or more slots therein, in accordance with an embodiment.

(39) FIG. 14B is a cross-section view of a portion of a dual O-ring rotary flow control valve assembly having a rotating drive shaft in the form of a cylindrical shaft configuration with one or more slots therein, in accordance with an embodiment.

(40) FIG. 14C is a top cross-section perspective view of the dual O-ring rotary flow control valve assembly with a rotating drive shaft having a cylindrical shape and one or more slots therein in a closed state configuration, in accordance with an embodiment.

(41) FIG. 14D is a top cross-section view of the dual O-ring rotary flow control valve assembly with a rotating drive shaft having a cylindrical shape and one or more slots therein in an open state configuration, in accordance with an embodiment.

(42) The drawings referred to in this description should be understood as not being drawn to scale except if specifically noted.

DESCRIPTION OF EMBODIMENTS

(43) The detailed description set forth below in connection with the appended drawings is intended as a description of various embodiments of the present invention and is not intended to represent the only embodiments in which the present invention is to be practiced. Each embodiment described in this disclosure is provided merely as an example or illustration of the present invention, and should not necessarily be construed as preferred or advantageous over other embodiments. In some instances, well known methods, procedures, and objects have not been described in detail as not to unnecessarily obscure aspects of the present disclosure.

Terminology

(44) In the following discussion, a number of terms and directional language is utilized. Although the technology described herein is useful on a number of vehicles that have an adjustable saddle, a bicycle will be used to provide guidance for the terms and directional language.

(45) In general, a bicycle has a front (e.g., the general location of the handlebars and the front wheel) and a rear (e.g., the general location of the rear wheel). For purposes of the discussion the front and rear of the bicycle can be considered to be in a first plane. A second plane that is perpendicular to the first plane would be similar to an exemplary flat plane of the ground upon which the bicycle is ridden.

(46) The term “seat tube” refers to a portion of a bicycle frame to which a seatpost is attached (often by insertion of a portion of the seatpost into the seat tube).

(47) A seatpost is a stand-alone component, e.g., a tube or other geometric shaped member, that couples a bicycle saddle with the bicycle frame via the seat tube. In one embodiment, the bottom of the seatpost is designed to be inserted into the seat tube of the bicycle frame and the top of the seatpost will include (or be coupled to) a saddle clamp assembly. The saddle clamp assembly is used to couple a bicycle saddle with the seatpost, in one embodiment, by clamping with the saddle rails.

(48) In assembly, the seatpost (with or without the saddle attached thereto) is partially inserted into the seat tube of the bicycle frame. In general, a user adjusts the amount of seatpost sticking out of the seat tube to establish the vertical height of the saddle (e.g., how far the saddle is above the ground plane, above the pedals, etc.). Once the seatpost (with saddle) is adjusted within the seat tube to obtain the desired saddle height and orientation, a clamping member (or another retaining device) is used about the seat tube to fasten the seatpost within the seat tube.

(49) The saddle clamp assembly allows a user to adjust the horizontal location of the saddle (e.g., toward the front or rear of the bicycle) and the pitch of the saddle (e.g., nose-up, nose-level, nose-down). In a standard seatpost, once the desired saddle height is established, the seatpost is clamped into position where it remains until it is unclamped. This singular saddle height capability is

important to allow different riders to utilize similar components and merely adjust the saddle height. However, as a rider tackles different challenges, it is becoming clear that a rider-to-bicycle geometry changes depending upon the terrain being traversed. For example, on a level road, the rider would have a certain saddle to pedal distance. However, when going down a hill (or over rough terrain, if standing for additional leverage, etc.), the same rider would likely prefer a shorter saddle to pedal distance to allow the rider to lower their center of gravity, lean further forward or backward, use their legs to absorb bumps, and the like. As such, it is helpful to be able to adjust the saddle height during a ride.

(50) A dropper seatpost assembly (hereinafter dropper seatpost) is a seatpost that includes a lower post, an upper post, and an actuator assembly. In the following discussion, the actuator assembly is a rotary flow control valve assembly.

(51) In one embodiment, the lower post is a hollow or semi-hollow design. In one embodiment, the upper post is a hollow or semi-hollow design. In one embodiment, the lower post and the upper post are telescopically coupled such that the overall length of the dropper seatpost is modified by adjusting the telescoping extension and retraction.

(52) In one embodiment, the telescoping extension and retraction capability of the upper and lower posts is controlled by the rotary flow control valve assembly. In one embodiment, the rotary flow control valve assembly is located in the lower post. In one embodiment, the rotary flow control valve assembly is located in the upper post. In one embodiment, the rotary flow control valve assembly could span the upper post and the lower post.

(53) Overview

(54) The following discussion discloses a rotary flow control valve assembly. In one embodiment, the rotary flow control valve assembly is an electronic rotary flow control valve assembly. In one embodiment, the rotary flow control valve assembly is a mechanically actuated rotary flow control valve assembly.

(55) In one embodiment, the rotary flow control valve assembly includes a wired communication and actuation capability. For example, in one embodiment, the rotary flow control valve assembly is used as an actuator in a dropper seatpost assembly where the drop function is actuated via a wired connection between the rotary flow control valve assembly and a user interface.

(56) In one embodiment, the rotary flow control valve assembly includes a wireless communication and actuation capability. For example, in one embodiment, the rotary flow control valve assembly is used as an actuator in a dropper seatpost assembly where the drop function is actuated via a wireless remote connection between the user interface and a motor used to rotate the rotary flow control valve of the rotary flow control valve assembly.

(57) In one embodiment, the dropper seatpost doesn't move under electrical power, but instead, the rotational opening or closing of the rotary flow control valve assembly used in the dropper seatpost assembly is what receives the signal and utilizes the electrical power. The actual compression of the dropper seatpost assembly is caused by the rider's body weight on the saddle and the return of the dropper seatpost assembly is provided by a spring return force (or the like).

(58) In one embodiment, using a communication protocol such as, but not limited to, those disclosed herein, the wirelessly actuated rotary flow control valve assembly used for the dropper seatpost assembly will respond to the remote input as fast or faster than a cable actuated dropper seatpost assembly. In other words, in one embodiment, the time lag, from the signal initiation by the rider using the wireless user interface until the wireless command is received and acted on by the electronic rotary flow control valve assembly causing the response in the dropper seatpost assembly, is smaller than a user perceptible delay.

(59) In one embodiment, using a communication protocol such as, but not limited to, those disclosed herein, the wired actuated rotary flow control valve assembly used for the dropper seatpost assembly will respond to the input from the user interface as fast or faster than a cable actuated dropper seatpost assembly

(60) In one embodiment, the rotary flow control valve assembly uses small and light componentry with a focus on both the minimizing of power requirements resulting in a long battery life and the minimizing of the weight/rotational inertia of the rotary flow control valve assembly. In one embodiment, such as in a dropper seatpost assembly, the packaging envelope for the rotary flow control valve assembly should be smaller than the diameter of the seat tube within which the dropper seatpost assembly is to be installed.

(61) In the following discussion, the operation of the rotary flow control valve assembly is provided in the context of a dropper seatpost assembly. However, in another embodiment, the rotary flow control valve assembly may be used in other active valve suspensions and components, to include other hydraulic applications such as a fork, shock, brake, etc. embodiments of different active valve suspension and components that may utilize the rotary flow control valve assembly are disclosed in U.S. Pat. Nos. 8,838,335; 9,353,818; 9,682,604; 9,797,467; 10,036,443; 10,415,662; the content of which are incorporated by reference herein, in their entirety.

(62) In one embodiment, the rotary flow control valve assembly and dropper seatpost assembly could be used on one or more of a variety of vehicles such as, but not limited to, a bicycle, an electric bike (e-bike), a moped, or the like. In one embodiment, when the rotary flow control valve assembly is used in a component other than a dropper seatpost assembly, the rotary flow control valve assembly could be used on a plurality of different vehicles, components, and the like. However, in the following discussion, and for purposes of clarity, a bicycle is utilized as the example vehicle.

(63) Operation

(64) Referring now to FIG. 1, a perspective view of a bicycle **50** is shown in accordance with an embodiment. In general, the bicycle **50** includes pedals, wheels, a chain or other drive mechanism, brakes, an optional suspension, a saddle **10**, a handlebars **200**, a dropper seatpost assembly **300**, a user interface **205**, and a bicycle frame **119**. In one embodiment, dropper seatpost assembly **300** is used to adjustably retain the saddle height and yaw position of saddle **10** with respect to bicycle frame **119**.

(65) In general, dropper seatpost assembly **300** includes an upper post, a lower post, and a rotary flow control valve assembly. The upper post and the lower post are telescopically coupled together to form the seatpost. In one embodiment, the upper post includes the saddle clamp assembly at a top thereof (e.g., at the end (or close to the end) of the upper post opposite the end of the upper post telescopically coupled with the lower post). In one embodiment, the lower post is inserted into and then fixedly coupleable with the seat tube **93** of bicycle frame **119**.

(66) In one embodiment, the rotary flow control valve assembly controls the telescoping capability of the upper post and lower post configuration, such that a user can operate a control lever (e.g., user interface **205** shown in FIG. 2) to “drop” the dropper seatpost assembly **300** to a lower setting (e.g., the saddle clamp assembly is approximately at the top of the lower post), and then use the same control lever to “return” the dropper seatpost assembly **300** to its preset ride height. This two-position capability allows a rider to have a preferred saddle ride height and also a lowered saddle height for traversing downhills, bumpy terrain, while standing on the pedals, or the like. Although two positions is discussed, the dropper seatpost assembly **300** could be adjustable to any number of different ride height positions, the use of two positions is discussed herein for purposes of clarity.

(67) In one embodiment, the amount that dropper seatpost assembly **300** extending from the bicycle frame **119** can be adjusted. In general, dropper seatpost assembly **300** may be made of various materials, such as, but not limited to: steel, aluminum, titanium, carbon fiber, and aluminum wrapped in carbon fiber. Further discussion of dropper seatpost assembly **300** is provided herein to include the discussion of FIGS. 3A-B.

(68) Referring now to FIG. 2, a perspective view of handlebar **200** having the user interface **205** coupled therewith is shown in accordance with an embodiment. In one embodiment, the user interface **205** is mounted on handlebar **200**. In one embodiment, user interface **205** is coupled with

handlebar **200** via a clip or other retaining device. In one embodiment, user interface **205** communicates seatpost height instructions for the dropper seatpost assembly **300** to rotary flow control valve assembly **333** via a wired connection, via a wireless connection, or via a combination of wired and wireless connections.

(69) In one embodiment, user interface **205** includes a wireless transmitter/receiver and is wirelessly coupled with rotary flow control valve assembly **333**. Of note, the user interface **205** may be, but is not limited to, any of the following components capable of wirelessly communicating with the dropper seatpost assembly **300**, e.g., a voice activation device, a GPS device having stored map, a smart phone, smart device, lever, button, or the like. Moreover, although the user interface **205** is shown coupled with handlebar **200**. In another embodiment, the user interface **205** could be located on another portion of the bicycle frame **119**, on a mount coupled with the vehicle, worn as a smart device, carried by the rider, or the like.

(70) In one embodiment, user interface **205** includes at least one control, such as the first user interface **205A** and may include a second user interface **205B**, it should be understood that in an embodiment, there may be only a single control, or in an embodiment there may be a set of controls. In one embodiment, when the cyclist interacts the user interface **205**, a signal is sent from the user interface **205** to the rotary flow control valve assembly **333**. As described in detail herein, the signal causes a rotation of a rotary flow control valve within the rotary flow control valve assembly **333**. The rotation of the rotary flow control valve causes the rotary flow control valve to open and/or close fluid flow between two or more fluid volumes within the dropper seatpost assembly **300**. This rotational opening and/or closing of the rotary flow control valve allows the dropper seatpost assembly **300** to drop to a lower saddle ride height, or return to a previous saddle ride height as discussed in further detail herein.

(71) With reference now to FIG. 3A, a perspective view of a dropper seatpost assembly **300** coupled with a saddle clamp assembly **350** is shown in accordance with an embodiment. In one embodiment, the dropper seatpost assembly **300** includes an upper post **310**, a lower post **315**, a rotary flow control valve assembly **333** and a bottom **326**. In one embodiment, some, part, or all of the rotary flow control valve assembly **333** is located in the lower post. In another embodiment, some, part, or all of the rotary flow control valve assembly **333** (shown as **333b** for purposes of clarity) is located in the upper post. In one embodiment, some, part, or all of the rotary flow control valve assembly **333** (shown as **333a** for purposes of clarity) could span the upper post and the lower post.

(72) In one embodiment, e.g., a gravel or road bicycle, the dropper seatpost assembly **300** travel does not need to be as long and as such, the lower post **315** of the dropper seatpost assembly **300** can be trimmed or otherwise shortened. Therefore, in one embodiment, the location of the rotary flow control valve assembly **333** could be placed further toward the saddle **10** within the dropper seatpost assembly **300**, such that an amount of material (e.g., a trimmable portion) could be removed from the outer post of the dropper seatpost assembly **300**. In one embodiment, the trimmable option may also be important for purposes of weight reduction, a better fit between dropper seatpost assembly **300** and seat tube **93** (and/or bicycle frame **119**), user preference, and the like.

(73) Although FIG. 3A shows a number of rotary flow control valve assembly **333** locations, in general, there is only one rotary flow control valve assembly **333** and the shown locations of rotary flow control valve assembly **333** (e.g., **333**, **333a**, **333b**, etc.) are indicative of a few of the possible placement locations for rotary flow control valve assembly **333**.

(74) In one embodiment, seat tube collar **325** is the highest portion of the lower post **315** and is indicative of the lowest possible setting for the dropper seatpost assembly **300** when it is installed into the bicycle frame **119** seat tube **93**.

(75) In one embodiment, the lower post **315** includes a top opening (e.g., approximately at seat tube collar **325**) to receive the upper post **310** and a tubular sidewall axially extending between the

top opening and the bottom **326** to form the lower post **315**, the outer diameter (OD) of the tubular sidewall of the lower post **315** is smaller than an inner diameter (ID) of a seat tube **93** of bicycle frame **119**, the lower post **315** for insertion into the seat tube **93**. In one embodiment, bottom **326** is the lowest portion of lower post **315** relative to when lower post **315** is within seat tube **93**.

(76) In one embodiment, upper post **310** telescopically slides with respect to lower post **315**. In one embodiment, the upper post **310** has an OD smaller than the ID of the lower post **315**, such that a portion of the upper post **310** can telescopically slide within the lower post **315**. In one embodiment, the upper post **310** has an ID larger than an OD of the lower post **315**, such that a portion of the lower post **315** can telescopically slide within the upper post **310**.

(77) In one embodiment, upper post **310** and at least part of saddle clamp assembly **350** are formed as a single component. In another embodiment, upper post **310** and saddle clamp assembly **350** consist of two or more distinct and/or different components.

(78) In one embodiment, when movement of the saddle is desired, (e.g., due to hills, terrain, aerodynamics, speed, etc.), a rider will cause the dropper seatpost assembly **300** to lower by triggering user interface **205** while the rider also depresses the saddle. In one embodiment, the user interface **205** will send a wireless signal to rotary flow control valve assembly **333** causing rotary flow control valve **420** to open a flow pathway such that the dropper seatpost assembly **300** will be capable of being moved down or up. In one embodiment, dropper seatpost assembly **300** has an air spring and use the rider's weight to move the saddle down, and will only raise the saddle back to the initial position when the rotary flow control valve assembly **333** is activated (e.g., wirelessly via user interface **205**). In one embodiment, dropper seatpost assembly **300** is "micro-adjustable". There are two types of micro-adjustable seatposts: (1) seatposts that can be continuously adjusted to an infinite number of positions; and (2) seatposts that can only be adjusted to a predetermined (preprogrammed) number of positions.

(79) For example, with regard to dropper seatposts that can only be adjusted to a preprogrammed number of positions, the dropper seatpost adjustment positions may be that of the following three positions: up; middle; and down. Generally, the rider prefers that the dropper seatpost assembly **300** be in the "up" position during a ride over flat terrain, a road surface, or pedaling up small hills on a road surface. The rider generally prefers that the dropper seatpost assembly **300** be in the "middle" position when the rider still wants a small amount of power through pedaling but yet would still like the saddle to be at least partially out of the way. This situation may occur while riding down a gentle hill or when the rider anticipates having to climb a hill immediately after a short decent. The rider generally prefers that the dropper seatpost assembly **300** be in the "down" position when the rider is standing up to provide the most amount of power through pedaling and wants the saddle to be at its lowest possible out of the way setting, when the rider is descending a hill, traversing bumpy terrain (e.g., bunny hopping, using flexed legs to absorb bumps, pump track type scenarios, etc.), or the like. For example, the lowest saddle position would be valuable during a decent where the rider would be positioned rearward of the saddle thereby moving the center of gravity lower and/or rearward resulting in a more stable and safer decent.

(80) Additional details regarding the operation of a dropper seatpost assembly is found in U.S. Pat. No. 9,422,018 entitled "Seatpost" which is assigned to the assignee of the present application, and which is incorporated herein by reference in its entirety.

(81) Referring now to FIG. **3B**, a cutaway view of the rotary flow control valve assembly **333** in the dropper seatpost assembly **300** of FIG. **3A** is shown in accordance with an embodiment. For purposes of clarity, a discussion of the components that were visible and/or described in FIG. **3A** will not be repeated herein, but are incorporated by the discussion of FIG. **3A** in their entirety.

(82) In FIG. **3B**, dropper seatpost assembly **300** includes IFP (Internal Floating Piston) **381**, translating shaft **402**, and cutaway sectional view **391**. In one embodiment, the IFP **381** charge allows a gas spring to be used to extend the dropper seatpost assembly **300**. In cutaway sectional view **391**, the rotary flow control valve assembly **333** is shown at the bottom **326** of dropper

seatpost assembly **300**. It should be appreciated that the cross section is used to show one embodiment of the configuration of dropper seatpost assembly **300** including the location of IFP **381** and the rotary flow control valve assembly **333**. However, as provided in further discussion herein, in another embodiment, one or more details of rotary flow control valve assembly **333** including different possible installation locations, variations, components, operational characteristics and the like are possible. The use of the embodiment of FIG. 3B is provided as an example of one embodiment and used herein for purposes of clarity.

(83) Electronic Rotary Flow Control Valve Assembly

(84) With reference now to FIG. 4A, a cutaway sectional view **391** (as identified in FIG. 3B) of a portion of the dropper seatpost assembly **300** including the rotary flow control valve assembly **333** is shown in accordance with an embodiment. In one embodiment, rotary flow control valve assembly **333** includes a piston **405** connected to a translating shaft **402**, a first chamber or inner chamber which is a fluid chamber pressurized by a rider's weight on the saddle **10**. For purposes of clarity, the first chamber is referred to hereinafter as an inner fluid chamber **410** (or inner pressure tube). In one embodiment, rotary flow control valve assembly **333** also includes a second chamber having an annular region about inner fluid chamber **410** which is pressurized on extension and by the IFP **381**. For purposes of clarity, the second chamber is referred to hereinafter as an outer fluid chamber **415** (or outer pressure tube). In one embodiment, the actions of the two chambers are reversed, e.g., the inner fluid chamber **410** is pressurized on extension and by the IFP **381** and the outer fluid chamber **415** is pressurized by the rider's weight on the saddle **10**.

(85) In one embodiment, rotary flow control valve assembly **333** is an electronic rotary flow control valve assembly which includes a rotary flow control valve **420** with a drive feature **425** which is coupled to motor **450**. In one embodiment, the rotational input from motor **450** into drive feature **425** will change the rotary position of rotary flow control valve **420**.

(86) In one embodiment, motor **450** is a brushed DC motor with a gearbox. In one embodiment, motor **450** is a stepper motor, brushless motor, coreless motor, or the like.

(87) In one embodiment, there is a cutout **422** in a portion of the rotary flow control valve **420** that interfaces with the lug **423** to create a hard stop. In one embodiment, the hard stop is used in the control system for the motor **450** as a current limit. For example, in one embodiment, when the motor **450** is activated, it will run until the motor **450** hits its current limit and is shut off. In so doing, the electronic rotary flow control valve assembly will quickly rotate the rotary flow control valve **420** the approximate 90 degree throw without requiring any additional controllers, inputs, etc.

(88) In one embodiment, the cutout **422** in a portion of the rotary flow control valve **420** that interfaces with the lug **423** is used to key the rotary flow control valve **420** to the cross holes **503**.

(89) In one embodiment, rotary flow control valve assembly **333** is (effectively) a two-state valve. In other words, the rotary flow control valve assembly **333** is an on/off valve. In one embodiment, the rotary flow control valve **33** is an on/off valve such that it is fast enough in its response such that a user would not be able to stop the dropper seatpost assembly **300** between states (e.g., state 1—the original user set saddle height and state-2 the lowest dropper seatpost setting).

(90) In one embodiment, the rotary flow control valve assembly **333** may have intermediate states (to limit flow, such as a high flow, a medium flow, a slow flow, etc., but not at zero flow). For example, the rotary flow control valve assembly **333** could have intermediate settings to control flow. In one embodiment, there may be a control system (an encoder on motor **450** with different settings thereon, a stepper motor, etc.) to control/adjust the on/off type rotating valve into one or more intermediate states, (e.g., between on and off), to provide a regulated flow.

(91) In general, translating shaft **402** moves up and down with the saddle **10**. In one embodiment, the translating shaft **402** is attached to the piston **405** to create the inner fluid chamber **410** which is at high pressure when the rider weight is being supported. In one embodiment, the outer fluid chamber **415** is a fluid chamber that is annular to the inner fluid chamber **410**. When the rotary

flow control valve **420** is opened, the fluid in the inner fluid chamber **410** moves through the open rotary flow control valve **420** and into the outer fluid chamber **415**. In one embodiment, the movement of the fluid into the outer fluid chamber **415** will move the IFP **381** upward toward the top of the dropper seatpost assembly **300** as the dropper seatpost assembly **300** is compressed (or moved into its lower height).

(92) Mechanically Actuated Rotary Flow Control Valve Assembly

(93) With reference now to FIG. **4B**, a cross-section view (identified in FIG. **3B**) of a portion of the dropper seatpost assembly **300** including a mechanical actuator **481** (e.g., a cable, hydraulic line, etc.) for mechanically actuating rotary flow control valve assembly **333** is shown in accordance with an embodiment. In general, the operation of the embodiment shown in FIG. **4B** is similar to that of FIG. **4A**, except for the change from a motor **450** used to electronically drive the rotary flow control valve assembly **333** of FIG. **4A**, to a mechanical actuator to mechanically actuate the rotary flow control valve assembly **333**.

(94) In one embodiment, the mechanical actuator **481** would provide a control capability such that a user input on a user interface **205** (or similar type device) would provide a mechanical actuation of the rotary flow control valve within the rotary flow control valve assembly **333** to change the state from open, to closed, and/or to partially open.

(95) With reference again to FIG. **3B** and FIGS. **4A** and **4B**, in one embodiment, the components of rotary flow control valve assembly **333** are coupled together during manufacture/assembly to form a single rotary flow control valve assembly **333**. In one embodiment, rotary flow control valve assembly **333** is broken down into two or more distinct and/or different component assemblies which are connectively coupled during installation to form an operational rotary flow control valve assembly **333**.

(96) For example, in one embodiment where the rotary flow control valve assembly **333** is broken down into two or more distinct and/or different component assemblies, one component assembly will include the piston **405**, translating shaft **402**, inner fluid chamber **410**, outer fluid chamber **415**, rotary flow control valve **420**, and part of drive feature **425**, and another component assembly will include part of a first assembly (e.g., may be located within the dropper seatpost assembly **300** and sealed from atmosphere while the drive feature **425** will extend through a seal of some type and out of the bottom **326** of dropper seatpost assembly **300**) and the motor **450** is installed at a different location on the bicycle **50**. Thus, upon installation of the dropper seatpost assembly **300** into the seat tube **93**, the rotary flow control valve assembly **333** will be operationally assembled when the drive feature **425** of the rotary flow control valve **420** is in mechanical contact with motor **450**.

(97) In one embodiment, this mechanical contact occur when the motor **450** is installed within the seat tube **93** and the drive feature **425** makes mechanical contact therewith. In another embodiment, this mechanical contact occurs when the motor **450** is installed somewhere else on the vehicle and a mechanical connection is made between the drive feature **425** and the motor **450**. In one embodiment, the mechanical connection may be a cable or the like that is coupled between the motor **450** and the drive feature **425**, thereby allowing the output of motor **450** to be rotationally transferred to drive feature **425**.

(98) In one embodiment, rotary flow control valve assembly **333** also includes a battery power source and a transmitter/receiver (described herein) to provide the input signal and power to motor **450** causing motor **450** to operate the rotary flow control valve **420**. In one embodiment, the battery power source is a disposable battery. In one embodiment, the battery is a rechargeable battery. In one embodiment, the battery can be recharged wired or wirelessly.

(99) Single Assembly Discussion

(100) In one embodiment, the components of rotary flow control valve assembly **333** are installed together during the build of dropper seatpost assembly **300**, such that the piston **405**, translating shaft **402**, inner fluid chamber **410**, outer fluid chamber **415**, rotary flow control valve **420**, drive feature **425** and motor **450** are within the sealed atmospheric environment of the dropper seatpost

assembly **300**. In the following discussion, this is referred to as a single rotary flow control valve assembly **333** housing embodiment.

(101) In one embodiment, the single rotary flow control valve assembly **333** housing embodiment may include an O-ring or other type of seal about the drive feature **425** between the rotary flow control valve **420** and motor **450** to divide the single rotary flow control valve assembly **333** into a “wet” side and a “dry” side. In one embodiment, the “wet” side components (e.g., piston **405**, translating shaft **402**, inner fluid chamber **410**, outer fluid chamber **415**, rotary flow control valve **420**) are within the area of, and exposed to the working fluid while the “dry” side components, (e.g., motor **450**, transmitters, battery, non-contact charging components, and the like), are separated from the working fluid. Although the above discussion includes a list of one embodiment of “wet” side and “dry” side components. It should be appreciated that one or more of the components of the single rotary flow control valve assembly **333** could be moved from the “wet” side and/or “dry” side.

(102) In one embodiment, when single rotary flow control valve assembly **333** housing embodiment is completely installed within the dropper seatpost assembly **300**, the seal would be located somewhere along the drive feature **425** such that the working fluid can use the flow paths, e.g., inner fluid chamber **410**, outer fluid chamber **415**, rotary flow control valve **420**, cross holes **503** (of FIGS. 5B and 5E), and the like, and flow can be controlled by the “wet” components, while the “dry” components will remain separate from the working fluid.

(103) In one embodiment, the seal is also a pressure type fluid seal such that the “wet” components would be in a pressurized environment, while the “dry” components would remain at atmosphere. In one embodiment, by utilizing a pressure type fluid seal, the battery for the rotary flow control valve assembly **333** could be accessible for battery replacement. For example, in the dropper seatpost assembly **300** embodiment, some, or a portion of the bottom **326** could be removable to provide access to the battery and then be reinstalled to provide a level of protection from debris, water, etc. to the dropper seatpost assembly **300**.

(104) In one embodiment, by utilizing a pressure type fluid seal, a charging port for the battery of the single rotary flow control valve assembly **333** housing could be provided in the dropper seatpost assembly **300** wherever the single rotary flow control valve assembly **333** housing is located. For example, when the single rotary flow control valve assembly **333** housing is located close to, or in proximity of the bottom **326**, bottom **326** could include a charging port. In one embodiment, the charging port would include a dust cover (or the like) to provide a level of protection from debris, water, etc. to the dropper seat post assembly when it is not in use.

(105) In one embodiment, e.g., the single rotary flow control valve assembly **333** housing embodiment, the battery (or other power source such as a capacitor, etc.) has a wirelessly rechargeable capability such that the battery could be charged using a wireless power transfer system. E.g., using an inductive charger (or the like) within a given distance of the wirelessly rechargeable capability of the battery.

(106) Wireless charging, in its most basic form utilizes a copper coil to create an oscillating magnetic field, which can create a current in one or more receiver antennas. In general, the wireless charger could be a charging pad that use tightly-coupled electromagnetic inductive or non-radiative charging; A charging bowl or through-surface type charger that uses loosely-coupled or radiative electromagnetic resonant charging to transmit a charge a few inches; An uncoupled radio frequency wireless charger that allows a trickle charging capability at distances of many feet, or the like.

(107) Examples of a wireless power transfer systems that could be used in one or more embodiments include those defined by the wireless power consortium (WPC) Qi standard, the AirFuel Alliance (e.g., Duracell Powermat, PowerKiss, etc.), WiTricity, and the like.

(108) In one embodiment, by using wireless power transfer, the battery can be charged even though it is sealed within the dropper seatpost assembly **300**. In one embodiment, the battery can be charged while the dropper seatpost assembly **300** is installed in the seat tube **93**. This can be

dependent upon factors such as, the bicycle frame **119** (e.g., composite, metal, thin, thick, etc.), the type of wireless power transfer being used, etc.

(109) In one embodiment, the dropper seatpost assembly **300** is removed from seat tube **93**, and the portion of the dropper seatpost assembly **300** containing the rotary flow control valve assembly **333**, including the battery, would be located proximate to the wireless charger to obtain the wireless charge.

(110) Electronic Rotary Flow Control Valve

(111) With reference now to FIG. 5A, a perspective view of a rotary flow control valve **420** is shown in accordance with an embodiment. In one embodiment, rotary flow control valve **420** includes at least one sealing portion **433** and one or more slots **431**. In general, when the at least one sealing portion **433** is aligned with the one or more cross holes **503** (shown in FIGS. 5B and 5D), fluid flow between inner fluid chamber **410** and outer fluid chamber **415** is stopped. In contrast, when one or more slots **431** are aligned with the one or more cross holes **503** (shown in FIGS. 5B and 5C), fluid flow between inner fluid chamber **410** and outer fluid chamber **415** can occur.

(112) In general, the direction of the fluid flow is based on the pressure differential between the working fluid within the inner fluid chamber **410** and outer fluid chamber **415**. For example, if the dropper seatpost assembly **300** is in its extended state and the rider is sitting on the saddle **10**, there will be more pressure on the fluid in inner fluid chamber **410**. As such, when rotary flow control valve **420** is opened, the working fluid will flow from inner fluid chamber **410** to outer fluid chamber **415**.

(113) In contrast, if the dropper seatpost assembly **300** is in its compressed state, the IFP **381** will be charged and there will be more pressure on the fluid in outer fluid chamber **415**. As such, when rotary flow control valve **420** is opened, as long as the rider is not sitting on the saddle **10**, the fluid will flow from outer fluid chamber **415** into inner fluid chamber **410** due to the existing pressure differential.

(114) In one embodiment, the rotary flow control valve **420** is a short throw valve where the difference between the open and the closed position is approximately 90 degrees or less. In one embodiment, the dropper seatpost assembly **300** will begin moving prior to the rotary flow control valve **420** completing its throw. For example, in one embodiment the exposed cross holes **503** have an angular sweep. As such, as soon as the rotary flow control valve **420** begins to expose the cross holes **503** between the inner fluid chamber **410** and the outer fluid chamber **415** fluid will start flowing and the dropper seatpost assembly **300** will start moving.

(115) In one embodiment, the rotary flow control valve **420** is designed such that the angular rotation between the sealed state and the start of flow is minimized. In one embodiment, instead of the cross holes **503** being circular, other shapes and combinations of shapes may be used between the body and the rotary flow control valve **420**. In one embodiment, instead of using single cross holes **503**, a grid of holes are used instead of single cross holes. In one embodiment, a grid of holes, horizontal slot, other shapes or combinations of shapes are used instead of single cross holes to reduce the opportunity for seal extrusion.

(116) In one embodiment, the rotary flow control valve **420** is a short throw valve where the difference between the open and the closed position is approximately 180 degrees or less.

(117) In one embodiment, the rotary flow control valve **420** is a single rotation valve where the difference between the open and the closed position is approximately 360 degrees or less.

(118) FIG. 5B is a partially cutaway view of the rotary flow control valve **420** of FIG. 5A installed within the rotary flow control valve assembly **333** with cross holes **503** and is shown in accordance with an embodiment. FIG. 5C, is a cross-section view of the rotary flow control valve **420** installed within the rotary flow control valve assembly **333** with cross holes **503** of FIG. 5B not shown in accordance with the position of rotary flow control valve **420** per an embodiment.

(119) In FIGS. 5B and 5C, the rotary flow control valve **420** has been rotated (as shown by

rotational arrow **512**) into an open position such that fluid may flow **510** between the inner fluid chamber **410** and the outer fluid chamber **415** (not shown in this Figure for clarity) via the one or more slots **431** in the rotary flow control valve and through cross holes **503**. For example, as discussed herein, when the saddle **10** is being compressed, the fluid will flow **510** from the inner fluid chamber **410** through the one or more slots **431** in rotary flow control valve **420** and cross holes **503** and into the outer fluid chamber **415**. In contrast, when the saddle **10** is returning to its ride height (e.g., the dropper seatpost assembly **300** is extending), the fluid will flow **510** from the outer fluid chamber **415** through the cross holes **503** and the one or more slots **431** in rotary flow control valve **420** and into the inner fluid chamber **410**.

(120) FIG. 5D is a partially cut-away view of the rotary flow control valve **420** of FIG. 5A installed within the rotary flow control valve assembly **333** in a closed position (e.g., blocking cross holes **503**) shown in accordance with an embodiment. FIG. 5E, is a cross-section view of the rotary flow control valve **420** installed within the rotary flow control valve assembly **333** in a closed position (e.g., the sealing portion **433** blocking fluid flow through cross holes **503**) shown in accordance with an embodiment.

(121) In FIGS. 5D and 5E, the rotary flow control valve **420** has been rotated (as shown by rotational arrow **513**) into a closed position such that sealing portion **433** is aligned with cross holes **503** and fluid cannot flow between the inner fluid chamber **410** and the outer fluid chamber **415** (not shown in this Figure for clarity). Although rotational arrows **512** and **513** are shown as being operated in a first direction for opening and then in a second direction for closing the rotary flow control valve **420**, one or more embodiments are well suited to other rotational directions, distances, etc.

(122) In one embodiment, the rotary flow control valve assembly **333** includes the rotary flow control valve **420** and a rotary motor **450**, such that all of the motion is rotary and as such, there is no need for any rotary to linear conversion. In other words, in one embodiment, there is no rotary to linear transmission and therefore no rotary to linear transmission is required. E.g., there is no need to convert the rotating motion from a motor to linear actuation motion.

(123) Therefore, as the rotary flow control valve assembly **333** relies only on rotational motion, no axial extension is needed for the components to which it is installed (e.g., dropper seatpost assembly **300**, or the like) other than the size of the rotary flow control valve assembly **333**. In one embodiment, to provide an even smaller axial footprint, the output shaft from motor **450** is used as the drive feature **425** for rotary flow control valve **420**.

(124) In one embodiment, the rotary flow control valve **420** has bi-directional sealing. In other words, the rotary flow control valve **420** will fluidly seal the inner fluid chamber **410** from fluid communication with the outer fluid chamber **415** when the dropper seatpost assembly **300** is in the extended state to support compressive forces, e.g., the rider's interactions with the saddle **10** while riding. In so doing, the rider can ride the bike without the dropper seatpost assembly **300** lowering down before rotary flow control valve assembly **333** receives a command to do so. Moreover, in one embodiment because of the bi-directional sealing, the rotary flow control valve **420** will also fluidly seal the inner fluid chamber **410** from fluid communication with the outer fluid chamber **415** when the dropper seatpost assembly **300** is in the lowered state to support extension forces, e.g., the saddle **10** will remain in the lowered position even when upward forces are applied to the saddle **10**. In so doing, the rider can use the saddle **10** (and or dropper seatpost assembly **300**) to lift up some or all of the bike without the dropper seatpost assembly **300** moving upward. In other words, dropper seatpost assembly **300** will remain in the lowered state until rotary flow control valve assembly **333** receives a command to rotate the rotary flow control valve **420** into an open (or partially open) valve position thereby allowing the dropper seatpost assembly **300** to extend.

(125) FIG. 6A is a perspective view of the head of rotary flow control valve **420** with one or more O-ring type seals **625** installed in a gland in accordance with an embodiment. FIG. 6B is a top view of the head of rotary flow control valve **420** with one or more O-ring type seals **625** installed in

glands thereon in accordance with an embodiment.

(126) FIG. 7A is a perspective cross-section view of the rotary flow control valve **420** with one or more preload pads **720** installed thereon in accordance with an embodiment. In one embodiment, each of the preload pads **720** consists of sealing material (e.g., a Teflon pad or the like) and a compliant material **725** is used to establish the preload on the sealing material of preload pads **720**. FIG. 7B is a perspective view of the rotary flow control valve **420** with one or more preload pads **720** installed thereon in accordance with an embodiment. FIG. 7C is a side view of the rotary flow control valve **420** with one or more preload pads **720** installed thereon in accordance with an embodiment.

(127) FIG. 8A is a top perspective cross-section view of the rotary flow control valve **420** with one or more preload pads **820** installed thereon in accordance with an embodiment. In one embodiment, each of the preload pads **820** consists of sealing material (e.g., a Teflon pad or the like) and a spring **825** is used to establish the preload on the preload pads **820**. FIG. 8B is a perspective view of the rotary flow control valve **420** of FIG. 8A with one or more preload pads **820** installed thereon in accordance with an embodiment.

(128) Rotary Flow Control Valve with Rotating Seal

(129) FIG. 9A is a perspective view of a rotary flow control valve **420** with an optional flow hole **905** therethrough, in accordance with an embodiment. In one embodiment, rotary flow control valve **420** of FIG. 9A is a self-charging seal design that includes an optional through hole (or flow hole **905**) through a portion of rotary flow control valve **420**, one or more slots **431**, a drive feature **425**, and at least one sealing portion **433**. In one embodiment, the optional flow hole **905** acts like a “charging port” to provide additional fluid pressure to sealing portion **433** to retain a zero fluid flow rate between inner fluid chamber **410** and outer fluid chamber **415**. In one embodiment, the flow hole **905** is optional depending upon the operating pressures on the fluids in one or both of inner fluid chamber **410** and/or outer fluid chamber **415**.

(130) Referring now to FIG. 9A and to FIGS. 4A and 4B, in one embodiment, the at least one sealing portion **433** includes a first seal portion **901** and a second seal portion **902**. In one embodiment, first seal portion **901** seals to an inner diameter of the cross holes **503** in the wall separating the inner fluid chamber **410** from the outer fluid chamber **415**. In contrast, second seal portion **902** provides an additional seal surface area that is provided against the area of the wall surrounding the cross holes **503**, and is used to increase the area of the at least one sealing portion **433** that is exposed to the fluid pressure differential (as shown in FIGS. 9B-9D).

(131) In one embodiment, the size of second seal portion **902** is tailored to the specific application in which the rotary flow control valve **420** and/or the rotary flow control valve assembly **333** is being used. For example, if the sealing area of sealing portion **433** was equivalent only to the first seal portion **901** (e.g., the bore or flow hole **905** diameter), the high pressure would act over the entire area and the friction would be high. However, by using the increased size of sealing portion **433** to include the second seal portion **902**, the area on both sides of the area that the high pressure is acting on are balanced, thereby limiting the normal force between the sealing portion **433** and the flow hole **905**.

(132) Similar to the discussion in the description of FIG. 5A, in FIG. 9A, when the at least one sealing portion **433** is aligned with the one or more cross holes **503** (shown in FIGS. 5B and 5D), fluid flow between inner fluid chamber **410** and outer fluid chamber **415** is stopped. In contrast, when one or more slots **431** are aligned with the one or more cross holes **503** (shown in FIGS. 5B and 5C), fluid flow between inner fluid chamber **410** and outer fluid chamber **415** can occur.

(133) FIG. 9B is a cross-section view of the rotary flow control valve **420** with a flow hole **905** therethrough installed within the rotary flow control valve assembly **333**, in accordance with an embodiment. In one embodiment, rotary flow control valve **420** of FIG. 9B is a self-charging seal design that includes a through hole (or flow hole **905**) through a portion of rotary flow control valve **420** and one or more plungers **920**.

(134) Referring now to FIG. 9B and to FIGS. 4A and 4B, in a compression state of the dropper seatpost assembly **300**, such as when the dropper seatpost assembly **300** is extended and the rider is sitting on the saddle **10**, the high pressure from the inner fluid chamber **410** passes through the flow hole **905** of the rotary flow control valve **420** (as indicated by arrow **910**) and increases a force to the one or more plungers **920** thereby increasing the sealing force keeping fluid from flowing out of inner fluid chamber **410** through the one or more cross holes **503** and into outer fluid chamber **415**.

(135) FIG. 9C is a top perspective cross-section view of a rotary flow control valve **420** with a flow hole **905** therethrough installed within the rotary flow control valve assembly **333**, in accordance with an embodiment. FIG. 9D is a cross-section view of the rotary flow control valve **420** with a flow hole **905** therethrough installed within the rotary flow control valve assembly **333**, in accordance with an embodiment. In one embodiment, rotary flow control valve **420** of FIGS. 9C and 9D is a self-charging seal design that includes a through hole (or flow hole **905**) through a portion of rotary flow control valve **420** and one or more plungers **920**.

(136) Referring now to FIGS. 9C, 9D, and to FIGS. 4A and 4B, in an extension state of the dropper seatpost assembly **300**, such as when the dropper seatpost assembly **300** is in its lowest height configuration (e.g., fully compressed), the high pressure is provided from the outer fluid chamber **415** and passes using a flow path indicated by arrows **935** along a portion of one or more plungers **920**. This high pressure feeds into the one or more O-rings **934** which increases a force applied from the one or more plungers **920** to the one or more sealing portions **433**. This increase in force, increases the sealing force of one or more plungers **920** which keeps fluid from flowing from outer fluid chamber **415** through the flow hole **905** and into inner fluid chamber **410**.

(137) FIG. 10A is a cross-section view of the rotary flow control valve **420** with a single O-ring installed within the rotary flow control valve assembly **333**, in accordance with an embodiment. In one embodiment, rotary flow control valve assembly **333** includes the rotary flow control valve **420** with drive feature **425**, inner fluid chamber **410**, at least one O-ring **1020**, a rod **1010**, and one or more seals **1033**.

(138) In one embodiment, rod **1010** is used to support the at least one O-ring **1020** so that it does not collapse under any high pressures from the working fluid. The at least one O-ring **1020** charges the one or more seals **1033** and there is some area between the one or more seals **1033** and the rotary flow control valve **420** that is charged by the high pressure fluid in inner fluid chamber **410** causing the one or more seals **1033** to move outward toward cross holes **503**.

(139) FIG. 10B is a perspective view of the rotary flow control valve **420** with a single O-ring configuration of FIG. 10A. FIG. 10B shows the rotary flow control valve **420**, the one or more seals **1033**, and the rod **1010**.

(140) FIG. 10C is a top perspective cut away view of the rotary flow control valve **420** with a single O-ring configuration installed within the rotary flow control valve assembly **333** in a closed state, in accordance with an embodiment. In the closed state, rod **1010** is used to support the at least one O-ring **1020** so that it does not collapse under any high pressures from the working fluid. The at least one O-ring **1020** charges the one or more seals **1033** and there is some area between the one or more seals **1033** and the rotary flow control valve **420** that is charged by the high pressure fluid in inner fluid chamber **410** causing the one or more seals **1033** to move outward toward cross holes **503** thereby closing the fluid path through cross holes **503**.

(141) FIG. 10D is a top perspective cut away view of the rotary flow control valve **420** with a single O-ring configuration installed within the rotary flow control valve assembly **333** in an open state, in accordance with an embodiment. In the open state, rod **1010** continues to support the at least one O-ring **1020** so that it does not collapse under any high pressures from the working fluid. The at least one O-ring **1020** charges the one or more seals **1033** and there is some area between the one or more seals **1033** and the rotary flow control valve **420** that is charged by the high pressure fluid in inner fluid chamber **410** causing the one or more seals **1033** to move outward. However, since the rotary flow control valve **420** is now rotated such that the one or more seals

1033 are moving toward the wall **1065** between the inner fluid chamber **410** and the outer fluid chamber **415** and not toward cross holes **503** in wall **1065**, the one or more slots **431** in rotary flow control valve **420** are providing an open fluid path between outer fluid chamber **415** and inner fluid chamber **410**. In so doing, the fluid can flow from the higher pressure differential to the lower pressure differential between inner fluid chamber **410** and outer fluid chamber **415** through one or more slots **431** and cross holes **503**.

(142) Rotary Flow Control Valve with Rotating Shaft

(143) FIG. **11A** is a cross-section view of a portion of a rotary flow control valve assembly **333** with a rotating shaft configuration, in accordance with an embodiment. In one embodiment, the reason for using a rotating shaft style versus a rotating seal style is that in the rotating seal style the friction interface (e.g., between the sealing portion **433** and the wall **1065** between inner fluid chamber **410** and outer fluid chamber **415**) is as far away from the centerline as possible (e.g., on the outside perimeter of rotary flow control valve **420**). This distance from centerline is the longest possible moment arm and as such would maximize the torque required to rotate the rotary flow control valve **420**.

(144) In contrast, in the rotating shaft style of rotary flow control valve assembly **333** shown in FIG. **11A**, the friction interface is moved closer to the center line thereby reducing the length of the moment arm and thus the torque required to rotate the rotating shaft as compared to the torque required to rotate the rotary flow control valve **420** in the rotating seal configuration. In one embodiment, the housing of the rotary flow control valve **420** does not rotate. Instead, the shaft within the rotary flow control valve **420** will rotate within the housing.

(145) Referring still to FIG. **11A**, in one embodiment, rotary flow control valve assembly **333** includes a rotary flow control valve **420** housing **1104**, one or more cross holes **503**, at least one O-ring **1120**, a stub shaft **1110**, at least one sealing portion **1133**, a rotating drive shaft **425** (in the form of a ball valve, or other geometric shape), flow holes **905**, a motor output shaft **1144**, and a flow path **1118** shown in the flow holes **905** between the inner fluid chamber **410** and outer fluid chamber **415**; where the inner fluid chamber **410** would be found above the opening at the top of the rotating drive shaft **425** and the outer fluid chamber **415** would be to the exterior of the one or more cross holes **503**.

(146) In one embodiment, the rotary flow control valve **420** housing **1104** does not rotate. Instead, the drive shaft **425** is the portion that rotates. In one embodiment, the at least one sealing portion **1133** (e.g., ball valve seats) is a dynamic seal in the sense that there is motion between the rotating drive shaft **425** and the at least one sealing portion **1133**.

(147) In one embodiment, the at least one O-ring **1120** seals to the one or more cross holes **503** and the inner fluid chamber **410** and charge the ball valve seat (e.g., the at least one sealing portion **1133**) to create the pressure for the at least one sealing portion **1133**. The stub shaft **1110** is used to keep the at least one O-ring **1120** from compressing on itself under pressure.

(148) In one embodiment, the motor output shaft **1144** is coupled with the drive shaft **425**. As stated herein, in one embodiment, the motor output shaft **1144** and the drive shaft **425** may be a single piece. In one embodiment, drive shaft **425** includes a ball valve type of shape and has a flow hole **905** therethrough.

(149) Referring now to FIG. **11B**, a top cross-section view of the rotary flow control valve assembly **333** with a rotating drive shaft in a closed state configuration is shown in accordance with an embodiment. In the closed state, housing **1104** does not rotate, stub shaft **1110** is used to support the at least one O-ring **1120** so that it does not collapse under any high pressures from the working fluid. The at least one O-ring **1120** charges the at least one sealing portion **1133**, and the rotating drive shaft **425** with flow hole **905** is rotated to a closed position where the flow hole **905** is not aligned with cross holes **503** thereby stopping the fluid from using the fluid path of flow hole **905** from moving between inner fluid chamber **410** and outer fluid chamber **415** (as shown in FIGS. **4A** and **4B**).

(150) FIG. 11C is a top cross-section view of the rotary flow control valve assembly 333 with a rotating drive shaft in an open state configuration, in accordance with an embodiment. In the open state, housing 1104 still does not rotate, stub shaft continues to support the at least one O-ring 1020 so that it does not collapse under any high pressures from the working fluid. The at least one O-ring 1020 charges the at least one sealing portion 1133 and the rotating drive shaft 425 with flow hole 905 is rotated to an open position. In the open position, the flow hole 905 within rotating drive shaft 425 provides an open fluid flow path between outer fluid chamber 415 and inner fluid chamber 410. In so doing, the fluid can flow from the higher pressure differential to the lower pressure differential between inner fluid chamber 410 and outer fluid chamber 415 through the flow hole 905 and cross holes 503.

(151) FIG. 11D is a cross-section view of a portion of a rotary flow control valve assembly 333 with a rotating shaft configuration having a one piece ball valve seat, in accordance with an embodiment. In general, the components and operation of FIG. 11D are similar to those of FIG. 11A. As such, the discussion of FIG. 11D will only include the differences between FIG. 11D and FIG. 11A.

(152) In one embodiment, instead of being a plurality of different components/pieces, the at least one sealing portion 1133 is combined with stub shaft 1110 in a single piece ball valve seat 1180.

(153) Referring now to FIG. 12A, a cross-section view of a portion of a rotary flow control valve assembly 333 having a rotating drive shaft in the form of a cylindrical shaft configuration is shown in accordance with an embodiment. In one embodiment, rotary flow control valve assembly 333 includes a rotary flow control valve 420 housing 1104, at least one O-ring 1120, a stub shaft 1110, at least one sealing portion 1133, a rotating drive shaft 425 (in the form of a cylindrical shaft, or other geometric shape), flow holes 905, and a flow path 1118 shown in the flow holes 905 between the inner fluid chamber 410 and outer fluid chamber 415; where the inner fluid chamber 410 would be found above the opening at the top of the rotating drive shaft 425 and the outer fluid chamber 415 would be to the exterior of the stub shaft 1110.

(154) In one embodiment, the rotary flow control valve 420 housing 1104 does not rotate. Instead, the drive shaft 425 is the portion that rotates. In one embodiment, the at least one sealing portion 1133 (e.g., valve seats) is a dynamic seal in the sense that there is motion between the rotating drive shaft 425 and the at least one sealing portion 1133.

(155) In one embodiment, the at least one O-ring 1120 seals to the one or more cross holes and the inner fluid chamber 410 and charge the valve seat (e.g., the at least one sealing portion 1133) to create the pressure for the at least one sealing portion 1133. The stub shaft 1110 is used to keep the at least one O-ring 1120 from compressing on itself under pressure.

(156) In one embodiment, the motor output shaft is coupled with the drive shaft 425. As stated herein, in one embodiment, the motor output shaft and the drive shaft 425 may be a single piece. In one embodiment, drive shaft 425 includes a cylindrical shape and has a flow hole 905 therethrough.

(157) Referring now to FIG. 12B, a top cross-section view of the rotary flow control valve assembly 333 with a rotating drive shaft 425 having a cylindrical shape is shown in a closed state configuration in accordance with an embodiment. In the closed state, housing 1104 does not rotate, stub shaft 1110 is used to support the at least one O-ring 1120 so that it does not collapse under any high pressures from the working fluid. The at least one O-ring 1120 charges the at least one sealing portion 1133, and the rotating drive shaft 425 with flow hole 905 is rotated to a closed position where the flow hole 905 is not aligned with cross holes 503 thereby stopping the fluid from using the fluid path of flow hole 905 from moving between inner fluid chamber 410 and outer fluid chamber 415 (as shown in FIGS. 4A and 4B).

(158) FIG. 12C is a top cross-section view of the rotary flow control valve assembly 333 with a rotating drive shaft having a cylindrical shape is shown in an open state configuration in accordance with an embodiment. In the open state, housing 1104 still does not rotate, stub shaft continues to support the at least one O-ring 1020 so that it does not collapse under any high

pressures from the working fluid. The at least one O-ring **1020** charges the at least one sealing portion **1133** and the rotating drive shaft **425** with flow hole **905** is rotated to an open position. In the open position, the flow hole **905** within rotating drive shaft **425** provides an open fluid flow path between outer fluid chamber **415** and inner fluid chamber **410**. In so doing, the fluid can flow from the higher pressure differential to the lower pressure differential between inner fluid chamber **410** and outer fluid chamber **415** through the flow hole **905** and cross holes **503**.

(159) Referring now to FIG. **13A**, a cross-section view of a portion of a dual O-ring rotary flow control valve assembly **333** having a rotating drive shaft **425** in the form of a cylindrical shaft configuration is shown in accordance with an embodiment. In one embodiment, rotary flow control valve assembly **333** is similar to the rotary flow control valve assembly **333** of FIG. **12A**, and includes a rotary flow control valve **420** housing **1104**, at least one dynamic O-ring **1120**, at least one static O-ring **1220**, and at least one component **1233**, a stub shaft **1110**, a rotating drive shaft **425** (in the form of a cylindrical shaft, or other geometric shape), flow holes **905**, and a flow path **1118** shown in the flow holes **905** between the inner fluid chamber **410** and outer fluid chamber **415**; where the inner fluid chamber **410** would be found above the opening at the top of the rotating drive shaft **425** and the outer fluid chamber **415** would be to the exterior of the stub shaft **1110**.

(160) In one embodiment, the rotary flow control valve **420** housing **1104** does not rotate. Instead, the drive shaft **425** is the portion that rotates. In one embodiment, the at least one static O-ring **1220** seals to the inner diameter (ID) of the inner fluid chamber **410** and loads through the at least one component **1233** which loads the at least one dynamic O-ring **1120** which creates the dynamic seal between the rotating drive shaft **425** and the seal.

(161) In one embodiment, the stub shaft **1110** is used to keep the at least one O-ring **1120** and **1220** from compressing on themselves under pressure. In one embodiment, the motor output shaft is coupled with the drive shaft **425**. As stated herein, in one embodiment, the motor output shaft and the drive shaft **425** may be a single piece. In one embodiment, drive shaft **425** includes a cylindrical shape and has a flow hole **905** therethrough.

(162) Referring now to FIG. **13B**, a top cross-section view of the dual O-ring rotary flow control valve assembly **333** with a rotating drive shaft **425** having a cylindrical shape is shown in a closed state configuration in accordance with an embodiment. In the closed state, housing **1104** does not rotate, stub shaft **1110** is used to support the at least one O-ring **1120** and **1220** so that they do not collapse under any high pressures from the working fluid. The at least one static O-ring **1220** seals to the ID of the inner fluid chamber **410** and loads through the at least one component **1233** which loads the at least one dynamic O-ring **1120** which creates the dynamic seal between the rotating drive shaft **425** and the seal, and the rotating drive shaft **425** with flow hole **905** is rotated to a closed position where the flow hole **905** is not aligned with cross holes **503** thereby stopping the fluid from using the fluid path of flow hole **905** from moving between inner fluid chamber **410** and outer fluid chamber **415** (as shown in FIGS. **4A** and **4B**).

(163) FIG. **13C** is a top cross-section view of the dual O-ring rotary flow control valve assembly **333** with a rotating drive shaft having a cylindrical shape is shown in an open state configuration in accordance with an embodiment. In the open state, housing **1104** still does not rotate, stub shaft continues to support the at least one O-ring **1120** and **1220** so that they do not collapse under any high pressures from the working fluid. The at least one static O-ring **1220** seals to the ID of the inner fluid chamber **410** and loads through the at least one component **1233** which loads the at least one dynamic O-ring **1120** which creates the dynamic seal between the rotating drive shaft **425** and the seal, and the rotating drive shaft **425** with flow hole **905** is rotated to an open position. In the open position, the flow hole **905** within rotating drive shaft **425** provides an open fluid flow path between outer fluid chamber **415** and inner fluid chamber **410**. In so doing, the fluid can flow from the higher pressure differential to the lower pressure differential between inner fluid chamber **410** and outer fluid chamber **415** through the flow hole **905** and cross holes **503**.

(164) FIG. **14A** is a perspective view of a rotating drive shaft **425** in the form of a cylindrical shaft

configuration with one or more slots **1431** therein. In one embodiment, unlike the configurations where there is one or more flow holes **905** therein, in one embodiment, the rotating drive shaft **425** utilizes one or more slots **1431**.

(165) Referring now to FIG. **14B**, a cross-section view of a portion of a dual O-ring rotary flow control valve assembly **333** having a rotating drive shaft **425** in the form of a cylindrical shaft configuration with one or more slots **1431** is shown in accordance with an embodiment. In one embodiment, rotary flow control valve assembly **333** is similar to the rotary flow control valve assembly **333** of FIG. **13A**, and includes a rotary flow control valve **420** housing **1104**, at least one dynamic O-ring **1120**, at least one static O-ring **1220**, and at least one component **1233**, a stub shaft **1110**, a rotating drive shaft **425** (in the form of a cylindrical shaft, or other geometric shape), one or more slots **1431**, and a flow path **1118** shown along one or more slots **1431** between the inner fluid chamber **410** and outer fluid chamber **415**; where the inner fluid chamber **410** would be found above the opening at the top of the rotating drive shaft **425** and the outer fluid chamber **415** would be to the exterior of the stub shaft **1110**.

(166) In one embodiment, the rotary flow control valve **420** housing **1104** does not rotate. Instead, the drive shaft **425** is the portion that rotates. In one embodiment, the at least one static O-ring **1220** seals to the inner diameter (ID) of the inner fluid chamber **410** and loads through the at least one component **1233** which loads the at least one dynamic O-ring **1120** which creates the dynamic seal between the rotating drive shaft **425** and the seal.

(167) In one embodiment, the stub shaft **1110** is used to keep the at least one O-ring **1120** and **1220** from compressing on themselves under pressure. In one embodiment, the motor output shaft is coupled with the drive shaft **425**. As stated herein, in one embodiment, the motor output shaft and the drive shaft **425** may be a single piece. In one embodiment, drive shaft **425** includes a cylindrical shape and has one or more slots **1431** therethrough.

(168) Referring now to FIG. **14C**, a top cross-section perspective view of the dual O-ring rotary flow control valve assembly **333** with a rotating drive shaft **425** having a cylindrical shape and one or more slots **1431** therein is shown in a closed state configuration in accordance with an embodiment. In the closed state, housing **1104** does not rotate, stub shaft **1110** is used to support the at least one O-ring **1120** and **1220** so that they do not collapse under any high pressures from the working fluid. The at least one static O-ring **1220** seals to the ID of the inner fluid chamber **410** and loads through the at least one component **1233** which loads the at least one dynamic O-ring **1120** which creates the dynamic seal between the rotating drive shaft **425** and the seal, and the rotating drive shaft **425** with one or more slots **1431** is rotated to a closed position such that the one or more slots **1431** are not aligned with cross holes **503** thereby stopping the fluid from using the fluid path of flow hole **905** from moving between inner fluid chamber **410** and outer fluid chamber **415** (as shown in FIGS. **4A** and **4B**).

(169) FIG. **14D** is a top cross-section view of the dual O-ring rotary flow control valve assembly **333** with a rotating drive shaft having a cylindrical shape and one or more slots **1431** is shown in an open state configuration in accordance with an embodiment. In the open state, housing **1104** still does not rotate, and stub shaft continues to support the at least one O-ring **1120** and **1220** so that they do not collapse under any high pressures from the working fluid. The at least one static O-ring **1220** seals to the ID of the inner fluid chamber **410** and loads through the at least one component **1233** which loads the at least one dynamic O-ring **1120** which creates the dynamic seal between the rotating drive shaft **425** and the seal, and the rotating drive shaft **425** with one or more slots **1431** is rotated to an open position. In the open position, the one or more slots **1431** of rotating drive shaft **425** provide an open fluid flow path between outer fluid chamber **415** and inner fluid chamber **410**. In so doing, the fluid can flow from the higher pressure differential to the lower pressure differential between inner fluid chamber **410** and outer fluid chamber **415** through the one or more slots **1431** and cross holes **503**.

(170) Communications Protocol

(171) In general, the goal of the disclosed communication protocol is low latency and long battery life. In one embodiment, the network implements the proprietary low-latency low-power radio protocol to provide an effective transport for communication between rotary flow control valve assembly **333** and user interface **205**.

(172) In one embodiment, a unique ID is used during the programming/pairing of the rotary flow control valve assembly **333** with the user interface **205**. In one embodiment, the unique ID is used by rotary flow control valve assembly **333** to identify a valid user interface **205**. In one embodiment, any transmitted signal includes a unique identifier (ID) that identifies the user interface **205** that broadcast the signal. Thus, even when a number of different user interfaces are operating in the same environment, the rotary flow control valve assembly **333** will be able to identify when the signal is sent from the appropriate user interface **205**.

(173) In one embodiment, the unique ID, and other data is stored in an erasable programmable read-only memory on a version of rotary flow control valve assembly **333**. In one embodiment, the memory can be written to with RF energy, NFC protocols, or the like. As such, the memory could be updated via user interface **205**, a mobile device, a laptop, or the like.

(174) In one embodiment, the wireless signal is a “telegram” or the like that includes the unique identifier (ID) that identifies the rotary flow control valve assembly **333** and/or the user interface **205** that broadcast the telegram signal. Thus, even when several electronic and/or mechanically actuated versions of rotary flow control valve assemblies and/or user interfaces are operating in the same environment, the telegram signal will identify which device sent the signal. Although the unique ID is used in one embodiment, in another embodiment, a different portion of the telegram signal is used to identify the transmitting device.

(175) In one embodiment, the telegram signal is sent via a radio frequency (RF) transmitter such as used in a wireless personal area network (WPAN), a low power network (LPN), Internet of things (IoT) connectivity, or the like. In one embodiment, the RF protocol could be, but is not limited to, Bluetooth, WiFi, Bluetooth Low Energy (BLE), near field communication (NFC), UHF radio signal, Worldwide Interoperability for Microwave Access (WiMax), industrial, scientific, and medical (ISM) band, IEEE 802.15.4 standard communicators, Zigbee, ANT, ISA100.11a (wireless systems for industrial automation: process control and related applications) wireless highway addressable remote transducer protocol (HART), MiWi, IPv6 over low-power wireless personal area networks (6LoWPAN), thread network protocol, subnetwork access protocol (SNAP), and the like.

(176) In one embodiment, the rotary flow control valve assembly **333** and/or the user interface **205** transmitter is powered by a momentary generator such as ZF electronics AFIG-0007 or the like. In one embodiment, if the IEEE 802.15.4 standard is utilized, the rotary flow control valve assembly **333** and/or the user interface **205** will include an IEEE 802.2 logical link control sublayer to receive and interpret the signal.

(177) In one embodiment, the communication capabilities of the rotary flow control valve assembly **333** and/or user interface **205** resemble a system such as the ZF Electronics AFIS-5003 with a SNAP transmitter, and the receiving device will have a universal asynchronous receiver/transmitter (UART) interface supporting RS-232 or RS-485 using TTL logic levels to receive and interpret the signal.

(178) In one embodiment, the rotary flow control valve assembly **333** and/or the user interface **205** will periodically send a heartbeat (e.g., check-in message), to inform the other of the rotary flow control valve assembly **333** and/or the user interface **205** that they are still active. In one embodiment, the heartbeat is sent at a 1 Hz communication rate. In one embodiment, the other of the rotary flow control valve assembly **333** and/or the user interface **205** that did not send the heartbeat will provide a response message to confirm that there is a wireless connection therebetween.

(179) In one embodiment, a timer is used by the rotary flow control valve assembly **333** and/or the

user interface **205** to count down a check-in or heartbeat time period. In one embodiment, the time period measured by the timer is preset by the manufacturer. In one embodiment, the time period measured by the timer is adjustably set by the manufacturer, by the user, by a mechanic, based on the vehicle location, terrain type, or the like.

(180) In one embodiment, when the timer expires, the heartbeat is sent. In one embodiment, once the wireless connection between the rotary flow control valve assembly **333** and the user interface **205** is confirmed, the timer will be restarted.

(181) In one embodiment, if there is no response to the heartbeat with a predefined period of time, another heartbeat will be sent. In one embodiment, if there is still no response received, an additional pre-defined number of heartbeat signals will be sent.

(182) In one embodiment, a microprocessor on the rotary flow control valve assembly **333** and/or the user interface **205** has a built-in radio capable of standard Bluetooth Low Energy and other communication as part of the ISM Band technologies. In one embodiment, the rotary flow control valve assembly **333** and/or the user interface **205** uses Enhanced Shockburst (ESB). In one embodiment, the microprocessor on the rotary flow control valve assembly **333** and/or the user interface **205** build in radio protocol could be, but is not limited to, WiFi, NFC, UHF radio signal, WiMax, ISM band, an IEEE 802.15.4 standard communicator, Zigbee, ANT, ISA100.11a, wireless HART protocol, MiWi, IPv6, 6LoWPAN, thread network protocol, SNAP, and the like.

(183) In one embodiment, user interface **205** has two wireless radios, e.g., radio **1** and radio **2**. In one embodiment, radio **1** handles wireless communication with peripherals where latency and/or battery life is not a concern. For example, radio **1** could communicate with devices such as other controllers, sensors, mobile devices, a power meter, or the like where ~200 msec latency is not a problem. In one embodiment, radio **2** handles the ESB communication and is dedicated to any low latency/low power devices such as rotary flow control valve assembly **333**.

(184) In one embodiment, having two operational radios is not that good from a battery life standpoint, but, in one embodiment, the user interface **205** houses a relatively large rechargeable battery. Thus, the expectation for the user interface **205** battery burn time is hours not months.

(185) In one embodiment, rotary flow control valve assembly **333** has a single wireless radio that uses ESB communication and is always listening for a message from the user interface **205**. In one embodiment, the amount of power draw from the “always-listening” ESB radio is minimal. However, the always-listening is key for low latency communication.

(186) For example, when the user interface **205** has a message to send to rotary flow control valve assembly **333**, it can be nearly instantaneously received by rotary flow control valve assembly **333** (or within 2 msec). In one embodiment, it can be received so quickly because the rotary flow control valve assembly **333** does not have to wake up and try to bond with the user interface **205** (a strategy normally employed to conserve battery life). It is already awake and ready.

(187) In one embodiment, the initial operation of the wireless communication protocol is to pair the rotary flow control valve assembly **333** to the user interface **205**. In one embodiment, the wireless communication pairing is made resistant against attempts made by unauthorized actors trying to attack and control the system by performing authentication and encryption between the wireless components. In general, examples of system attacks include, but are not limited to, replay attacks, impersonation, denial of service, and the like.

(188) In one embodiment, replay attacks refers to actions such as, but not limited to, an attacker recording one or more of the messages and playing them back to the device which mistakenly interprets them as valid messages from the sensor(s).

(189) In one embodiment, impersonation refers to actions such as, but not limited to, an attacker pretending to be a sensor, and sending one or more messages directly to the rotary flow control valve assembly **333** and/or the user interface **205**.

(190) In one embodiment, denial of service refers to actions such as, but not limited to, an attacker sending one or more specially crafted messages that stop the system from working. Although a

number of examples of system attacks are discussed herein, the examples are not exhaustive. In contrast, it is possible, and should be appreciated, that other types of system/communication attacks may be utilized.

(191) In one embodiment, the authentication and encryption between the rotary flow control valve assembly **333** and/or the user interface **205** includes the utilization of AES 128, or the like. For example, in one embodiment, the pairing procedure sets up all state required for the radio protocol to be secure, including the AES-128 symmetric key. Whenever a device—the rotary flow control valve assembly **333** and/or the user interface **205**—is turned on, it generates a session-specific 4-byte nonce using a secure random number generator. This nonce is included in all communication between devices.

(192) Within a single session, each device also stores a 4-byte sequence number, that starts at 0, and increments for every transmitted message. The AES-128 block cipher is operated in the Authenticated Encryption with Associated Data (AEAD) scheme, which allows encrypting the given plaintext, and authenticating associated plain text data. The AEAD scheme requires a 13-byte nonce value, referred to herein as AEADNonce. When the AES-128 symmetric key, and AEADNonce are unique for every packet, the connection is secured.

(193) In one embodiment, the AEADNonce is constructed by concatenating the nonce of each device with the sequence number of the particular packet, for a total of 12 bytes, with the 13th byte padded with 00. This ensures the AEADNonce is unique, and the connection is therefore secure. In one embodiment, the application does not accept any packet which it receives that has a sequence number earlier than another packet it has already received. This ensures that replay attacks are not possible. To generate new packets with a valid sequence number, the attacker must know the AES-128 symmetric key.

(194) In one embodiment, to perform this activity, Bluetooth communication (or the like) is used. In one embodiment, to perform this activity, ESB type communications is used. In one embodiment, once the pairing is completed and each component (e.g., the rotary flow control valve assembly **333** and/or the user interface **205**) has the keys, any further communications are made via ESB for the fast-communication state. In other words, once the rotary flow control valve assembly **333** is connected to the user interface **205** the ESB protocol (or similar ISM Band Technology) takes over. For example, as discussed herein, the rotary flow control valve assembly **333** and/or the user interface **205** periodically send a heartbeat message (e.g., a check-in) to the other of the rotary flow control valve assembly **333** and/or the user interface **205** to make sure the system is working properly.

(195) In one embodiment, since the rotary flow control valve assembly **333** is always listening for a message from the user interface **205**, the rotary flow control valve assembly **333** will receive the message at the speed of the message being sent. In one embodiment, rotary flow control valve assembly **333** will send an acknowledgement message back to the user interface **205**. In one embodiment, rotary flow control valve assembly **333** will not send an acknowledgement message back to the user interface **205**.

(196) In one embodiment, the communication latency is approximately 2.2 milliseconds. Where, the Accelerometer i2C at 100 kHz is approximately 1000 microseconds, the encryption is approximately 610 microseconds, and the time from the initiation of the transmission from user interface **205** until usable data is received at rotary flow control valve assembly **333** (includes radio time and decryption) is approximately 545 microseconds.

(197) Sometimes a message will not be received properly by the rotary flow control valve assembly **333** and an associated acknowledgement message will not be returned to the user interface **205**. In one embodiment, a message resend is then initiated. Each message resend attempt adds an average of 0.5 milliseconds latency. For example, if user interface **205** does not receive an acknowledgement message from the rotary flow control valve assembly **333** within a given amount of time, the user interface **205** sends the message again. For example, assume a 10% chance of

message failure. In this case, there is a 10% chance of needing a second message attempt, a 1% chance of needing a 3rd message attempt, and a 0.1% chance of needing a 4th message attempt. Each message attempt adds about 0.5 milliseconds latency, so even a 4th message attempt will be within the “non-user perceptible” latency period.

(198) In one embodiment, one or both of the rotary flow control valve assembly **333** and/or the user interface **205** can be in a number of different energy states to conserve battery life. Although a number of states are discussed, in one embodiment there may be more, fewer, or a different combination or variation of the described energy states. The use of the disclosed energy states is provided herein as one embodiment and for purposes of clarity.

(199) One state is referred to as the operating state. This is the highest battery power consumption state. In the operating state, the rotary flow control valve assembly **333** and/or the user interface **205** is transmitting and/or receiving data.

(200) In a standby state, the rotary flow control valve assembly **333** and/or the user interface **205** are awake and there is a connection therebetween. In the standby state, the user interface **205** is waiting to receive input from the user. When the user provides an input, user interface **205** will move into the operating state and transmit the data to rotary flow control valve assembly **333**.

(201) In one embodiment, when rotary flow control valve assembly **333** responds to the transmission from user interface **205**, it will be known to both devices that there is a connection therebetween, that the signal has been received by rotary flow control valve assembly **333**, and that one or both the rotary flow control valve assembly **333** and/or the user interface **205** can return to the standby state until the next time the user provides an input to user interface **205**.

(202) In one embodiment, the rotary flow control valve assembly **333** may not provide a response to the transmission from user interface **205**. In one embodiment, user interface **205** may not expect a response from rotary flow control valve assembly **333** after user interface **205** sends the transmission.

(203) In one embodiment, the rotary flow control valve assembly **333** may only provide the heartbeat message to the user interface **205** at pre-defined intervals to evidence the connection between user interface **205** and rotary flow control valve assembly **333**.

(204) In one embodiment, if user interface **205** expected but did not receive a response from rotary flow control valve assembly **333**, user interface **205** will include a programmed pre-defined number of attempts at transmitting the signal to rotary flow control valve assembly **333** before making the determination that there is a disconnection in the communication between the user interface **205** and the rotary flow control valve assembly **333**.

(205) In one embodiment, the pre-defined number of attempts is based on the transmission rate. For example, in one embodiment, it is assumed that it takes 0.5 milliseconds of time for the user interface **205** to transmit a signal to rotary flow control valve assembly **333**, for rotary flow control valve assembly **333** to transmit a message received transmission to user interface **205** and also cause motor **450** to rotate rotary flow control valve **420** to an open position thereby activating the dropper seatpost assembly **300**.

(206) If the time between the user input occurring (e.g., the user inputting a dropper command to user interface **205**) and the movement of the dropper seatpost assembly **300** being noticed by the rider is 1.5 milliseconds, then the pre-defined number of attempts taken by user interface **205** would be 3. Thus, user interface **205** will have tried as many times as possible (e.g., 0.5×3) to send the signal before the lack of dropper seatpost assembly **300** movement was noticed by the rider.

(207) In one embodiment, the number of times the user interface **205** tries could be more or less. The use of three attempts is used herein in one embodiment.

(208) In one embodiment, such as after a period of inaction, or the user interface **205** determines that the bike is not being ridden (e.g., based on a user input, a sensor input, or the like) the user interface **205** will send a standby message to inform rotary flow control valve assembly **333** that the bike is not being ridden. In one embodiment, the rotary flow control valve assembly **333** will

transition to a low-power mode or sleep mode when the standby message is received.

(209) In one embodiment, once the rotary flow control valve assembly **333** determines user interface **205** is turned off (or otherwise not responding), rotary flow control valve assembly **333** will enter a no-heartbeat standby state (e.g., an intermediate battery power consumption state), where rotary flow control valve assembly **333** is awake and listening but is not sending any transmissions (e.g., heartbeat transmissions, etc.).

(210) In one embodiment, rotary flow control valve assembly **333** will remain in the no-heartbeat standby state until the connection with user interface **205** is re-established. In one embodiment, when rotary flow control valve assembly **333** receives a message from user interface **205**, it will know that the connection with user interface **205** is established (or re-established) and rotary flow control valve assembly **333** will transition from the no-heartbeat standby state to a heartbeat standby state.

(211) In dormant state, the bike is stationary. For instance, the bike is in storage or otherwise parked and not being ridden. In one embodiment, when in dormant state, one or both of the rotary flow control valve assembly **333** and/or the user interface **205** will go into low-power mode. In one embodiment, while in the dormant state, rotary flow control valve assembly **333** will periodically wake up to transmit a signal to user interface **205**. If no response is received, rotary flow control valve assembly **333** will return to the dormant state, e.g., go back to sleep.

(212) In contrast, if rotary flow control valve assembly **333** receives a response from user interface **205** during the periodic wakeup, in one embodiment, rotary flow control valve assembly **333** will change from the dormant state into the standby state.

(213) Thus, in one embodiment, the rotary flow control valve assembly **333** and/or the user interface **205** can move between the different states fluidly using the model described above. In one embodiment, the rotary flow control valve assembly **333** and/or the user interface **205** will try to remain in (or return to) the lowest powered state for the specific situation.

(214) In one embodiment, based on the different states described above, the estimate of battery life for a battery used by the rotary flow control valve assembly **333** and the user interface **205** is determined using a duty cycle such as, for example, 2 hours per ride, 4 rides per week, 48 weeks per year.

(215) In one embodiment, the power draw for each state is approximated as an average of 150 microamp draw during the active state, an average of 32 microamp draw during either standby state, and an average of 1 microamp draw during the dormant state.

(216) In one embodiment, the battery is a CR2032 battery. In a CR2032 battery, the capacity is approximately 173 mAh. As such, and based on the power draw for each state and the duty cycle example above, the expected battery life of the battery **339** of the rotary flow control valve assembly **333** and/or the user interface **205** is 23 months. In one embodiment, if the duty cycle is different, the lifespan of the battery will be different.

(217) In one embodiment, a different battery with a different capacity can be used. For example, a smaller (or lighter) battery, a rechargeable battery, or the like. For example, a road bike rider may want a smaller (or lighter) rotary flow control valve assembly **333** as the reduction of weight is one of the most important goals. As such, the rider could use an rotary flow control valve assembly **333** with a smaller battery and therefore swap the battery life (e.g., reduce the battery life from 23 months to a lower life span e.g., a few months, weeks, or the like), in order to obtain a weight savings.

(218) In contrast, an avid mount bike rider may have a harder duty cycle, with more time of the rotary flow control valve assembly **333** and/or the user interface **205** being in operating state, as such, the expected battery life would be reduced by the actual operation of the rotary flow control valve assembly **333** and/or the user interface **205**.

(219) The foregoing Description of Embodiments is not intended to be exhaustive or to limit the embodiments to the precise form described. Instead, example embodiments in this Description of

Embodiments have been presented in order to enable persons of skill in the art to make and use embodiments of the described subject matter. Moreover, various embodiments have been described in various combinations. However, any two or more embodiments can be combined. Although some embodiments have been described in a language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed by way of illustration and as example forms of implementing the claims and their equivalents.

Claims

1. A rotary flow control valve comprising: a drive mechanism; and a rotary flow control valve directly coupled with said drive mechanism, wherein said rotary flow control valve is rotated by said drive mechanism without requiring conversion of a linear motion to a rotational motion, said rotary flow control valve comprising: a substantially cylindrical body comprising: at least one slot, wherein said at least one slot allows for fluid flow when aligned with a cross hole; and at least one sealing portion, wherein said at least one sealing portion prevents fluid flow when aligned with said cross hole, wherein said at least one slot is at a particular angle from said at least one sealing portion, wherein said at least one slot is a first slot and a second slot, wherein said second slot is located 180 degrees from said first slot such that there is a first sealing portion that is located 180 degrees from a second sealing portion; a first preload pad disposed on said first sealing portion, wherein said first preload pad prevents fluid from leaking past said first sealing portion when said first sealing portion is aligned with said cross hole; a second preload pad disposed on said second sealing portion, wherein said second preload pad prevents fluid from leaking past said second sealing portion when said second sealing portion is aligned with a second cross hole, wherein said first preload pad and said second preload pad are made of a sealing material; and a spring disposed between said first preload pad and said second preload pad, said spring used to establish a preload on said first preload pad and said second preload pad.
 2. The rotary flow control valve of claim 1 wherein, there is no fluid flow path that goes through said substantially cylindrical body.
 3. The rotary flow control valve of claim 1 further comprising: an O-ring seal disposed in a gland on said sealing portion, wherein said O-ring seal prevents fluid from leaking past said sealing portion when said sealing portion is aligned with said cross hole.
 4. The rotary flow control valve of claim 1 further comprising: at least one preload pad located on said sealing portion comprising: a sealing material to prevent fluid from leaking past said sealing portion when said sealing portion is aligned with said cross hole; and a compliant material layered behind said sealing material, wherein said compliant material is used to establish a preload on said sealing material.
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